

# 43\_Croatia

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## 1 How have food imports evolved relative to fuel imports as a share of merchandise imports in Croatia between 1992 and 2024?

### 1.1 Abstract

Using World Bank World Development Indicators (WDI), this study examines the evolution of Croatia's import composition between 1992 and 2024. The analysis compares two key indicators: food imports (% of merchandise imports) and fuel imports (% of merchandise imports). Over this thirty-two-year period, food imports increased slightly and steadily, reflecting a moderate but consistent rise in the share of food products within Croatia's total merchandise imports. Fuel imports, by contrast, displayed a highly volatile trajectory: they rose sharply until 2012, then dropped dramatically to levels below those of 1992, followed by a significant rebound after 2020 and a moderate decline toward the end of the period. These patterns reveal the uneven dynamics of Croatia's import dependence, highlighting structural differences between essential goods like food and energy commodities such as fuel, and illustrating the economy's responsiveness to external shocks, global price fluctuations, and policy changes.

### 1.2 1. Question

How have food imports evolved relative to fuel imports as a share of merchandise imports in Croatia between 1992 and 2024?

- **Food imports proxy:** Food imports (% of merchandise imports)
- **Fuel imports proxy:** Fuel imports (% of merchandise imports)

### 1.3 2. Data

- **Source:** World Bank World Development Indicators (WDI)
- **Indicators:**
  - Food imports (% of merchandise imports)
  - Fuel imports (% of merchandise imports)
- **Coverage:** Croatia, 1992–2024
- **Notes:** National-level data only

### 1.4 3. Method

1. Filtered dataset for Croatia and selected the two import indicators.
2. **Extracted relevant columns:** Year, Indicator Name, and Value.

3. Pivoted the dataset to create a side-by-side chronological comparison of food and fuel imports as shares of merchandise imports.
4. Produced a dual-line time series plot to visualize trends, volatility, and long-term patterns in Croatia's import structure.

(Analysis is descriptive; no causal inference applied.)

## 1.5 4. Results

- **Food imports (% of merchandise imports):** Increased slightly and steadily over the period, indicating a moderate rise in the relative importance of food products in Croatia's imports.
- **Fuel imports (% of merchandise imports):** Showed significant volatility, rising sharply until 2012, then dropping dramatically to below 1992 levels, rebounding strongly after 2020, and declining moderately toward 2024.
- **Comparison:** The two indicators exhibit divergent patterns: food imports demonstrate stability with gradual growth, while fuel imports are highly sensitive to external factors, reflecting Croatia's exposure to global energy markets and price shocks.

(Figure 1. Croatia: Food vs. Fuel Imports as Share of Merchandise Imports, 1992–2024)

(Table 1. Pivoted dataset summary)

## 1.6 5. Interpretation

- The gradual increase in food imports suggests steady demand for essential goods, consistent with population needs and stable consumption patterns.
- The dramatic fluctuations in fuel imports reflect external shocks, such as global oil price volatility, supply disruptions, and energy policy changes.
- The divergence between the two indicators highlights the structural differences in Croatia's import portfolio, with food providing stability and fuel imports contributing substantial volatility to trade dynamics.
- Understanding these patterns is crucial for trade and energy policy, revealing the need to balance essential consumption stability with energy security and diversification.

## 1.7 6. Limitations

- National-level aggregates may conceal sectoral or regional import variations.
- WDI data rely on national reporting and modeled estimates, potentially introducing uncertainty, especially in earlier years.
- The descriptive approach does not isolate causal drivers, such as changes in energy infrastructure, trade agreements, or global commodity price shocks.

## 1.8 7. Next Steps / Extensions

- Analyze fuel imports by energy type (oil, gas, coal) to understand drivers of volatility.
- Examine correlations between food and fuel imports with GDP growth, domestic production, and energy consumption.
- Compare Croatia's import trends with neighboring EU countries to evaluate regional dependencies and resilience.

- Conduct time-series decomposition to distinguish long-term structural trends from short-term shocks in import patterns.

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[1]: # How have food imports evolved relative to fuel imports as a share of
      ↪merchandise imports in Croatia between 1992 and 2024?

import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "private-sector_hrv_filtered.csv" # Filtered dataset with only
      ↪relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").
      ↪reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Croatia dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
plt.figure(figsize=(10,6))
plt.plot(df_plot["Year"], df_plot["Food imports (% of merchandise imports)"],
      marker='o', linestyle='-', label="Food imports (% of merchandise
      ↪imports)")
plt.plot(df_plot["Year"], df_plot["Fuel imports (% of merchandise imports)"],
      marker='o', linestyle='-', label="Fuel imports (% of merchandise
      ↪imports)")
```

```
plt.title("Croatia: Food vs Fuel imports (% of merchandise imports) \
↳(1992-2024)")
plt.xlabel("Year")
plt.ylabel("Percentage")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder, "croatia_food_vs_fuel_imports.png"))
plt.show()

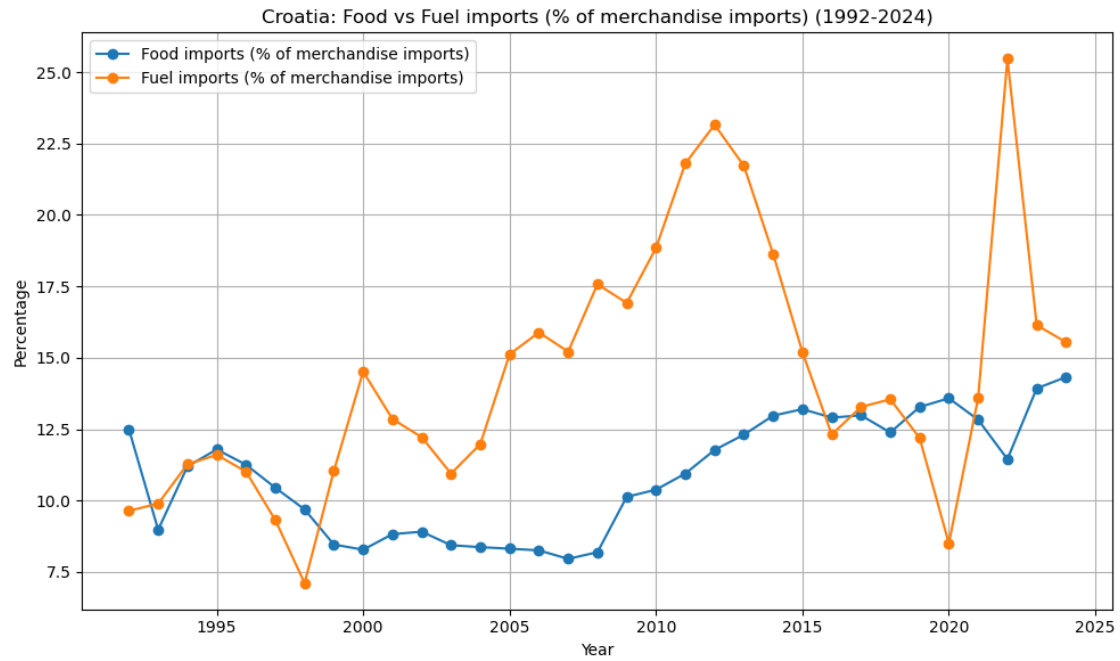
# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder, \
↳"croatia_food_vs_fuel_imports"), index=False)
```

Pivoted Croatia dataset:

| Indicator Name | Year | Food imports (% of merchandise imports) \ |
|----------------|------|-------------------------------------------|
| 0              | 1992 | 12.499903                                 |
| 1              | 1993 | 8.977340                                  |
| 2              | 1994 | 11.192911                                 |
| 3              | 1995 | 11.789644                                 |
| 4              | 1996 | 11.253315                                 |
| 5              | 1997 | 10.447293                                 |
| 6              | 1998 | 9.687793                                  |
| 7              | 1999 | 8.460247                                  |
| 8              | 2000 | 8.277969                                  |
| 9              | 2001 | 8.823567                                  |
| 10             | 2002 | 8.909018                                  |
| 11             | 2003 | 8.435962                                  |
| 12             | 2004 | 8.367178                                  |
| 13             | 2005 | 8.313900                                  |
| 14             | 2006 | 8.255557                                  |
| 15             | 2007 | 7.958334                                  |
| 16             | 2008 | 8.190360                                  |
| 17             | 2009 | 10.128611                                 |
| 18             | 2010 | 10.379410                                 |
| 19             | 2011 | 10.943771                                 |
| 20             | 2012 | 11.769570                                 |
| 21             | 2013 | 12.304679                                 |
| 22             | 2014 | 12.971060                                 |
| 23             | 2015 | 13.199163                                 |
| 24             | 2016 | 12.902358                                 |
| 25             | 2017 | 12.982256                                 |
| 26             | 2018 | 12.381682                                 |
| 27             | 2019 | 13.274568                                 |
| 28             | 2020 | 13.579731                                 |
| 29             | 2021 | 12.827479                                 |

|    |      |           |
|----|------|-----------|
| 30 | 2022 | 11.448594 |
| 31 | 2023 | 13.925556 |
| 32 | 2024 | 14.321451 |

| Indicator Name | Fuel imports (% of merchandise imports) |
|----------------|-----------------------------------------|
| 0              | 9.638816                                |
| 1              | 9.892011                                |
| 2              | 11.268764                               |
| 3              | 11.600026                               |
| 4              | 11.010440                               |
| 5              | 9.316662                                |
| 6              | 7.109416                                |
| 7              | 11.039330                               |
| 8              | 14.512213                               |
| 9              | 12.852310                               |
| 10             | 12.217404                               |
| 11             | 10.941926                               |
| 12             | 11.976959                               |
| 13             | 15.118617                               |
| 14             | 15.886516                               |
| 15             | 15.211403                               |
| 16             | 17.571375                               |
| 17             | 16.917283                               |
| 18             | 18.843362                               |
| 19             | 21.795942                               |
| 20             | 23.156740                               |
| 21             | 21.718243                               |
| 22             | 18.628321                               |
| 23             | 15.195733                               |
| 24             | 12.310865                               |
| 25             | 13.274770                               |
| 26             | 13.544163                               |
| 27             | 12.192121                               |
| 28             | 8.498070                                |
| 29             | 13.591754                               |
| 30             | 25.465091                               |
| 31             | 16.140514                               |
| 32             | 15.543368                               |



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